

Amjad S Althabteh

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Summary

C++ graphics and engine developer focused on real-time rendering and performance-critical systems. Built modular OpenGL engines, GPU-optimized pipelines, and custom allocators. Experienced with Unreal Engine 5 gameplay systems and low-level debugging.

Education

B.S in Computer Science	GPA: 3.65/4.00	Wayne State University	Detroit, MI	2026
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Experience

Graphics Engineer, Co-Founder	Immersalab Graphics	Detroit, MI	Sep 2025 - Dec 2025
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- Architected a C++20/OpenGL real-time rendering engine with GPU-driven pipelines and frame-time stability optimizations.
- Built internal UI tools and a JSON asset pipeline for flexible scene configuration.
- Implemented data-driven rendering optimizations for balanced GPU workloads.
- Developed GPU profiling and frame-timing tools for performance analysis.

Unreal C++ Dev	Indie Game (NOA)	Ann Arbor, MI	Apr 2025 - Aug 2025
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- Developed gameplay systems in C++20 and Unreal Blueprints integrating animation and narrative logic.
- Created and rigged 3D environments and characters.
- Produced a 90-second real-time cinematic using MetaHuman, Control Rig, and Sequencer.
- Built animation control systems using C++ and Animation Blueprints for smooth blending.

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Software Development Intern	Rare Munchies	Ann Arbor, MI	Sep 2023 - Apr 2024
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- Integrated UI components using **TypeScript**, improving page transitions and interaction flow.
- Identified and resolved **cross-browser compatibility issues**, reducing debugging time and unblocking delivery
- Optimized client-side rendering logic and state management patterns to reduce unexpected UI re-renders.

Research Experience

Undergraduate Researcher – Orbital Collision Modeling	Wayne State University—Detroit, MI Feb 2025 – Present
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- Modeled satellite trajectories using numerical integration and N-body gravitational dynamics.
- Simulated **orbital collision probabilities** using spatial partitioning techniques.

Projects

Glint 3D – C++ Automated 3D Rendering Engine	glint-rendering
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- Optimized a C++17/OpenGL rendering engine with improved GPU workload distribution and lighting systems.
- Added frustum culling and instanced rendering to reduce draw calls in large scenes.
- Implemented a custom C++20 memory pool allocator reducing heap allocations and runtime overhead.

3D Space Simulation Engine – C++/OpenGL	space-simulations
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- Built a 3D space simulation engine in C++17 using OpenGL, GLAD, and SFML.
- Designed modular engine systems for rendering, physics, math, and entity management.
- Implemented a custom math library (Vec3, Mat4) and camera system with projection/view matrices.

HFT OrderBook Engine — C++20

- Architected a price-time FIFO order matching engine with bid/ask depth tracking and partial fills.
- Designed low-latency data structures using cache-aligned memory and modern C++20 features.

C++ Runtime Stack Trace & Diagnostics Engine (Traceforge)

- Built a C++23 debugging system for automated stack-trace parsing and runtime diagnostics.
- Implemented automated regression test generation with intelligent fix suggestions.
- Added memory tracking utilities to detect leaks and invalid allocations.

Skills

- **Programming:** C++ (C++11–C++23), C, C#, Python, TypeScript, Java
- **Graphics & Engines:** OpenGL, Unreal Engine 4–5, Unity, Godot
- **Technical Areas:** Real-Time Rendering, Engine Architecture, Simulation Systems, GPU Optimization, Tools Development